Veer Chheda

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Education

Georgia Institute of Technology **Aug 2026 – May 2028**
Masters of Science in Computer Science

Dwarkadas J. Sanghvi College of Engineering **Nov 2022 – Jun 2026**
B.Tech in Computer Science and Engineering (IoT and Cybersecurity with Blockchain Technology) *CGPA 9.23/10 - Top 10*

Experience

Indian Institute of Technology (IIT) Patna **Aug 2024 – Nov 2025**
Research Intern *Remote*

- Conducting research under **Dr. Rajiv Misra** and**Dr. Sourav Dandapat**, focusing on NLP and computer vision.
- Developed mechanisms to identify and mitigate **hallucinations** in large language models for **MedQA** datasets using **ground-truth similarity** and **knowledge-base injection** for mitigation.
- Trained and optimized custom **NanoDet** and **YOLO** models for robust **small object detection** under varied conditions.
- Fine-tuned **LlaVA-next-video, Qwen 2.5 VL, Phi-4 mini** on multimodal summarization of car crash videos using both textual and visual data.

Aura-AI **Jun 2024 – Oct 2024**
Data Science Intern *Remote*

- Curated footfall analytics solutions, including **heatmaps** and **re-identification** using YOLO, achieving 86% accuracy in identifying customer retention patterns and hotspots, boosting brand visibility by 25%.
- Integrated **OCR** models using **Amazon Textract** into commercial pipelines for **automated data entry**, reducing manual processing time by 47%.
- Collaborated with a **cross-functional team** of 5+ members to design and deploy APIs using **FastAPI** and secure authentication using **JWT**.

Research and Publications

The Geometry of Personality in Language Models *First Author | Accepted*

- Accepted at **the Findings of IJCNLP-AACL 2026**. **Mechanistic interpretability** study of how the **OCEAN** personality traits are geometrically encoded across **5 models and their instruction-tuned variants** using the **Big5-Chat** dataset.
- Showed via **layer-wise probing** and **prequential MDL** that personality features compress monotonically with depth and that instruction tuning acts as a representational regulariser, but that trait directions remain in **overlapping subspaces** whose entanglement predicts cross-trait leakage under naive steering.
- Introduced **Null-Space Projection** (NSP), an inference-time method that orthogonalizes the target trait direction against competing trait subspaces, improving target shift by **25.7%** (LLM judge) and **35.7%** (human raters) while reducing cross-trait leakage by **31.9%**.

Extract-Explain-Abstract: A Rhetorical Role-Driven Domain-Specific Summarisation Framework for Indian Legal Documents *First Author | Paper*

- Published at the **7th Natural Legal Language Processing Workshop**, **EMNLP 2025**.
- Developed the **Extract-Explain-Abstract** (EEA) framework for summarizing Indian legal documents via rhetorical role segmentation, modeling logical dependencies between roles with chain-of-thought to cut **hallucinations by 17%**.
- Evaluated on the **ILC** and **IN-Abs** datasets with ROUGE, BERTScore, InLegalBERTScore, Flesch Reading Ease and SummaC.
- Human and expert evaluation showed **33%** preference and **40%** gold-standard replacement endorsement over baselines.

Iterative Critique-Driven Text Simplification: Targeted Enhancement of Complex Definitions with Small Language Models *First Author | Paper*

- Published at the **Findings of IJCNLP-AACL 2025**.
- Proposed an iterative zero-shot critique-driven framework for simplifying domain-specific definitions with **small language models**, using LLM critics to refine definitions for lay audiences across clarity, precision, conciseness and relevance.

- Benchmarked on the **WIKIDOMAINS** dataset (**22k+ definitions**), showing gains in BLEU-4, BERTScore, Age-of-Acquisition and Flesch Reading Ease.
- Human evaluation (**18 annotators**) and **LLM-as-a-judge** confirmed significant comprehension gains, especially for non-expert readers.

Could you BE more sarcastic? A Cognitive Approach to Bidirectional Sarcasm Understanding in Language Models *First Author* | Paper

- Published at the **Student Research Workshop**, **IJCNLP-AACL 2025**.
- Introduced a **multi-hop chain-of-thought** framework for bidirectional sarcasm style transfer, leveraging **Theory of Mind**.
- Benchmarked **6 small language models** on sarcasm generation and removal using the **MUStARD** dataset (690 dialogues), evaluated with automatic metrics, LLM-as-a-judge and human judgements.
- Improved metrics by up to **12%** over few-shot baselines and ranked best across all four style directions in human evaluation.

Color Space Fusion for Semantic Segmentation in Indian Adverse Weather Driving Scenes *First Author* | *Accepted*

- **Best Paper Award**, **6th Biennial International Conference on Nascent Technologies in Engineering**; to appear in **IEEE Xplore**. Benchmarked four color space fusion methods for RGB-NIR semantic segmentation across rain, fog, low-light and snow.

Evaluating Linguistic Proximity of Marathi as a Bridge Language for Konkani Translation *First Author* | *Accepted*

- **6th Biennial International Conference on Nascent Technologies in Engineering**; to appear in **IEEE Xplore**. Evaluated Marathi as a bridge language for English-to-Konkani translation using IndicTrans2, best among Devanagari-script Indo-Aryan languages.

Predicting Personality from Text: A Dual-Task Psycholinguistic Analysis of Machine Learning and Transformer Models *Second Author* | *Under review*

- Under review at the **8th IEEE International Conference on Emerging Smart Computing and Informatics**. Dual-task MBTI prediction with ensemble ML and transformer models, interpreted via **SHAP** and the **Empath** lexicon.

Projects

Compression-Aware LLM Inference Server | *PyTorch, FastAPI, Hugging Face Transformers* **Github**

- Built an LLM inference server with naive single-request serving, **continuous batching** with a per-sequence **KV-cache**, **INT8 post-training quantization** and **structured depth pruning**.
- Designed a **faithfulness/hallucination eval** over TruthfulQA, TriviaQA and CNN-DailyMail, scored 0–3 by an **LLM-as-a-judge** groundedness rubric, reporting **bootstrap 95% CIs** and judge reliability via **Cohen's kappa**.
- Ranked layers by **Block Influence** (ShortGPT) using forward hooks over a task-calibrated set; showed weight-only INT8 is near-lossless (perplexity **13.75** vs **13.76** baseline) while depth pruning degrades non-linearly (**37.11** at a 25% ratio) on Qwen2.5-0.5B-Instruct.

Vanilla-ML: Machine Learning from Scratch | *Python, NumPy* **Github**

- Implementing ML algorithms from first principles in **pure NumPy**, **hand-deriving every gradient** and documenting the underlying math.
- Built **linear and logistic regression, KNN, SVM, decision trees, k-means** and **PCA**, backed by a supporting math library of activations (ReLU, LeakyReLU, sigmoid, tanh, softmax), losses, metrics and optimizers (**SGD, AdaGrad, RMSProp, Adam**).
- Currently building a **scalar autograd engine** with a computation graph and hand-implemented backpropagation, working toward neural network layers and transformer blocks.

JainGPT - AI chatbot for Jainism | *Google Cloud, LangChain, ChromaDB* **Github**

- Designed a RAG application for custom knowledge chatbot using over 20 scriptures and books on Jainism.
- Combined ChromaDB and VertexAI to generate and store vector embeddings.
- Programmed prompt engineering for text retrieval and text translation into Gujarati, text-to-speech using Google Cloud APIs.

Technical Skills

Languages: Python, C++, C, JavaScript, Solidity, SQL, HTML/CSS, LaTeX
ML & NLP: PyTorch, Transformers, scikit-learn, Unsloth, LangChain, OpenCV, NumPy, Pandas, Matplotlib
Serving & MLOps: FastAPI, Uvicorn, Pydantic, Flask, Streamlit, pytest, Ruff, GitHub Actions, Git
Cloud, Data & Web: Google Cloud (Vertex AI), AWS (Textract), Firebase, MySQL, ChromaDB, React.js, Next.js, Vite, Tailwind CSS, Node.js, Hardhat, ethers.js